Fiscal Year:	FY 2018	**Task Last Updated:**	FY 03/26/2018
PI Name:	Chang, Polly Y. Ph.D.		
Project Title:	Particle Radiation-Induced Dysregulation of Protein Homeostasis in the Brain		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	polly.chang@sri.com	**Fax:**	FY 650-859-2889
PI Organization Type:	NON-PROFIT	**Phone:**	650-859-2549
Organization Name:	SRI International		
PI Address 1:	Biosciences, PN 187		
PI Address 2:	333 Ravenswood Avenue		
PI Web Page:			
City:	Menlo Park	**State:**	CA
Zip Code:	94025-3453	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	05/01/2014	**End Date:**	04/30/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 4/30/2019 per NSSC information (Ed., 4/11/18) NOTE: Change in period of performance (original period of performance was 2/26/2014-2/25/2018) per PI and NSSC information (Ed., 12/9/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lin, Hua Ph.D. (SRI International) Shaler, Thomas Ph.D. (SRI International)		
Grant/Contract No.:	NNX14AE69G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space atmosphere is comprised of wide spectrum of highly energetic and charged particle radiation. Exposure to this type of radiation in the galacto-cosmic environment or unplanned solar particle events (SPE) may pose a potential health hazard to humans in long-term manned space explorations. In normal biological systems, protein stability and system homeostasis are maintained through a tightly coupled post-translational modification process that is catalyzed by the ubiquitin-proteasome system for protein turnover. This process is necessary for modifying the correct protein in the correct cellular location at the correct time to control important processes, e.g., cell-cycle regulation, the maintenance of genomic stability, removal of damaged or misfolded proteins, autophagy, apoptosis, and gene transcription. Depending on the radiation dose and beam quality, particle radiation may cause significant damage to proteins and oxidative stress in the cells in the central nervous system, leading to dysfunction of the cellular ubiquitin-related pathways, an increase in the levels of extracellular ubiquitin, and persistent subsequent downstream effects on regulation of stress-associated signaling pathways in many different cell types within the central nervous system (CNS), precipitating neurodegenerative disease. We hypothesize that quantitative changes in cellular ubiquitin pools occur when neuronal cells are exposed to particle radiation and aim to study the impact of protein degradation after particle radiation using both in vitro and in vivo model systems. Information gathered from this research will aid in reducing the uncertainties associated with predicting risks both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate any potential negative effects in space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We expect that results associated with the regulation of protein processing and degradation in brain tissues obtained from our studies will be useful to understand molecular mechanisms of action in normal neuronal cell responses in particle radiation clinical therapy scenarios.
Task Progress:	Studies aimed at assessing the risk of neurodegeneration from exposure to the reactive oxygen species produced by HZE (high energy) radiation resulted in long-term production of oxidative stress. Among the molecular hallmarks of several neurodegenerative diseases, those manifesting from dysfunction of protein homeostasis in neurons resulting in the accumulation of ubiquitin-positive protein aggregates, are particularly prominent. The goal of our project is to further investigate radiation-quality dependent signaling mechanisms associated with protein homeostasis in brain tissues. Protein modification by ubiquitin is a highly regulated process catalyzed by enzyme families which can modify the correct protein in the correct cellular location at the correct time to control important processes such as cell cycle progression, removal of damaged or mis-folded proteins, and gene transcription. Currently, seven lysine residues have been identified as known points of ubiquitination, of which linkages at Lysine-11, -48, and -63 are the most abundant sites. Lysine 48-linked (designated as K-48) polyubiquitin chains target proteins for destruction by proteolysis. Lys-63-linked (also designated K-63) ubiquitination serves as a molecular platform for protein/protein interaction important for DNA damage repair, receptor endocytosis, kinase signaling activation, and protein trafficking. There is some evidence in the literature suggesting that atypical chains linked by Lys-11 can induce proteasomal degradation. Utility of these markers is evidence by analytical results of gray matter in brain tissues from patients who died of Alzheimer disease where significantly increased K48, 11, and 63 linkages that correlated with increased Braak staging of disease severity. Justified by its emerging role in aging related CNS diseases, the goal of our project is to dissect the unique role of these targeted protein degradation systems in particle radiation responses by quantitatively assess particle-radiation induced cellular protein turnover to uncover alterations in pathway-specific cellular ubiquitin pools and protein modification in vitro using cultured primary human and mouse neuronal stem cells, in vivo by evaluating particle radiation induced alterations in protein ubiquitination in brain tissues of C57Bl6 mice. To date, we have exposed human Neuronal Stem Cells to single doses of 250 MeV protons, 260 MeV/u Silicon, or 1 GeV/u Iron ions and harvested protein extracts from these cells ~24 h post irradiation for pUb analysis. No significant changes relative to control were observed in either K48 or K11 chain linkages after proton- or silicon-irradiation while significant linkage-dependent changes were observed after a single 50 cGy dose of Iron ions. Cultured mouse neuronal cells from three different regions in the brain: cortical, striatal, and mesencephalic showed that mouse striatal neurospheres were affected after particle radiation exposure. We noted dose- and linkage-dependent changes after either Silicon or Iron ion exposures. Hippocampal tissues from mice that were either sham-irradiated or total-body irradiated with protons, silicon, or iron ions using the same energies as the ones for cells described above were isolated up to 11 weeks post-irradiation and analyzed. Analysis of protein fractions showed no significant changes in pUb types after proton or silicon irradiation, independent of dose or time of tissue harvest. On the other hand, we noted the K48 linkages were elevated and K63-linkages were depressed at 11 wks after a single 100 cGy exposure of 1 GeV/u iron ions. Given that K-48 linkages are known to be associated with protein degradation via the ubiquitination enzymatic process and that K63 is the platform associated with DNA damage and repair, these results suggest that heavy ion induced changes in protein degradation in the CNS is a dynamic process where long-term consequences can arise many months after the initial insult. Using non-biased quantitative LC/MS/MS, greater than 1000 proteins were identified with altered expression in the mouse hippocampus tissue after iron or proton irradiation. Quantitative comparison of two candidate proteins, VGF neuro-peptide and ß-2 adaptin in mouse hippocampal tissue after Fe or proton exposure showed up-regulation after Fe at 11 wks post irradiation but not after protons at 8 wks post irradiation suggesting that radiation quality plays a role in temporal responsive protein regulation in the CNS.
Bibliography Type:	Description: (Last Updated: 04/26/2023)
Abstracts for Journals and Proceedings	Chang PY, Shaler TA, Lin H. "Particle-radiation induced dysregulation of protein homeostasis in the brain." Presented at 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. Proceedings. 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. , Oct-2017

Abstracts for Journals and Proceedings	Chang PY, Shaler TA, Lin H. "Particle-radiation induced dysregulation of protein homeostasis in the brain- an update." Presented at the 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018